

Multiple Choice (10 marks)

1. In Python 3, let `r = lambda q: q * 2`. What is `r(3)`?
 - (a) 2
 - (b) 6
 - (c) 3
 - (d) 1
2. Which of the following statements about a remote procedure call is true?
 - (a) It is used to call procedures with addresses that are farther than 2^{16} bytes away.
 - (b) It cannot return a value.
 - (c) It cannot pass parameters by reference.
 - (d) It cannot call procedures implemented in a different language.
3. In Python 3, which of the following function checks in a string that all characters are in uppercase?
 - (a) `isupper()`
 - (b) `join(seq)`
 - (c) `len(string)`
 - (d) `ljust(width[, fillchar])`
4. What is the value of this python expression: `4 + 3 % 2`?
 - (a) 1
 - (b) 6
 - (c) 4
 - (d) 5
5. In python 3, which of the following is floor division?
 - (a) `/`
 - (b) `//`
 - (c) `%`
 - (d) `|`

True / False (10 marks)

Answer True or False

6. True or False: Variable names in Python are not case-sensitive and can be used interchangeably regardless of their casing.
7. True or False: In a complete K-ary tree of depth N, the ratio of nonterminal nodes to total nodes is approximately $1/K$.
8. True or False: The minimum value of the list $l = [1,2,3,4]$ in Python 3 is 2.
9. True or False: The string 'xxzy' can be generated by the grammar defined by the rules $A \rightarrow BC$, $B \rightarrow x|Bx$, $C \rightarrow B|D$, $D \rightarrow y|Ey$, $E \rightarrow z$.
10. True or False: The value of the Python expression $1 + 3 \% 3$ is 1.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to divide two lists using map and lambda function.
12. Write a function to sort a list of tuples in increasing order by the last element in each tuple.

End of Paper